

Math Thermo  
class 01  
01/20/2026

## Non-Equilibrium Thermodynamics

### Course Plan

- 1) Equilibrium Thermodynamics
- 2) Statistical Mechanics
- 3) Kinetic Theory of Gases
- 4) Conservation laws in Continuous Systems
- 5) Entropy Production
- 6) Onsager's Principle
- 7) Applications

## Equilibrium Thermodynamics

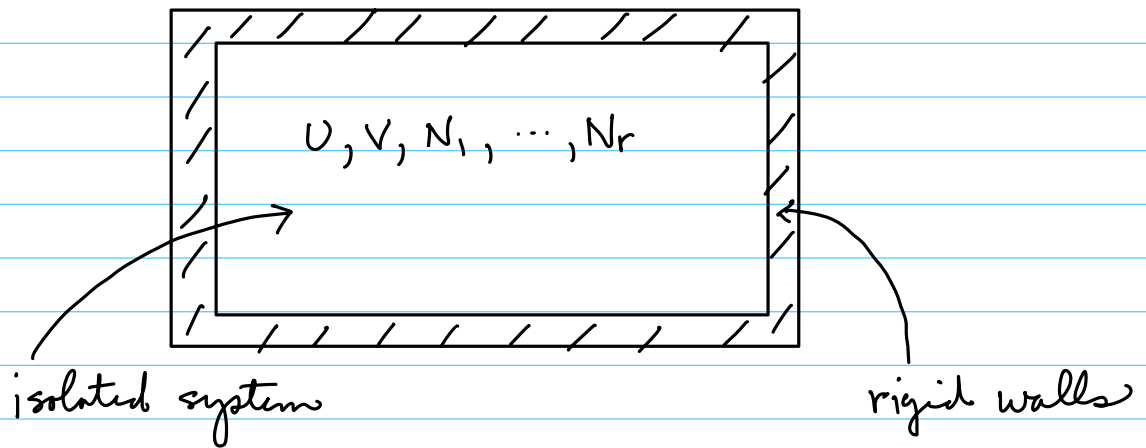
Defn (1.1): An isolated system is a collection of matter that shares no information with the outside world.

Postulate I: There are particular states, called equilibrium states, of an isolated system that, macroscopically, are characterized by

$U$ , the internal energy of the system,  $U \geq 0$ ,

$V$ , the volume of the system,  $V \geq 0$ ,

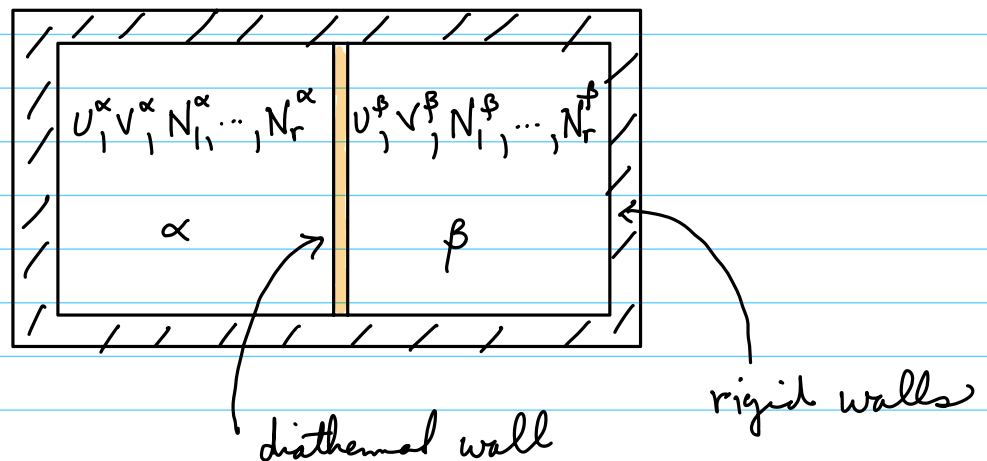
$N_1, \dots, N_r$ , the mole numbers of the  $r$  chemical components of the system,  $N_i \geq 0$ .



The rigid walls that isolate the system from the rest of the universe allow no change in volume, no exchange of matter, and no exchange of energy.

Defn (1.2): A composite system is the union of two isolated systems that can exchange volume, matter (chemical components), and/or energy, however, no volume, matter, or energy is exchanged with outside.

A diathermal wall in a composite system is one which separates two otherwise isolated systems and allows for the exchange of energy, but not matter or volume.



A diathermal piston in a composite system is one which separates two otherwise isolated systems and allows for the exchange of energy and volume but not matter.

Postulate II: There exists a function  $\tilde{S}$  for an isolated system called its entropy, defined for the system at equilibrium and depending on  $U, V, N_1, \dots, N_r$ , ie,

$$\tilde{S} = \tilde{S}(U, V, N_1, \dots, N_r).$$

The entropy exists for a composite system and is the sum of the entropy functions,  $\tilde{S}^\alpha$  and  $\tilde{S}^\beta$ , for the respective subsystems, ie,

$$(1.1) \quad \tilde{S} = \tilde{S}^\alpha + \tilde{S}^\beta = \tilde{S}^\alpha(U^\alpha, V^\alpha, N_1^\alpha, \dots, N_r^\alpha) + \tilde{S}^\beta(U^\beta, V^\beta, N_1^\beta, \dots, N_r^\beta)$$

The entropy of an isolated system is homogeneous of degree 1, meaning,

$$\tilde{S}(\lambda U, \lambda V, \lambda N_1, \dots, \lambda N_r) = \lambda \tilde{S}(U, V, N_1, \dots, N_r)$$

for any  $\lambda > 0$ .

Postulate III: The entropy of an isolated system is a concave, twice continuously differentiable, positive function over its (convex) domain

$$\Sigma_U \subseteq [0, \infty)^{r+2} = [0, \infty) \times [0, \infty) \times [0, \infty) \times \dots \times [0, \infty)$$

(U)            (V)            (N<sub>1</sub>)    ...    (N<sub>r</sub>)

The entropy is a monotonically increasing function of  $U$ . In particular,

$$\left(\frac{\partial \tilde{S}}{\partial U}\right)_{V, N_1, \dots, N_r} > 0, \quad \forall (U, V, \vec{N}) \in \overset{\text{interior of } \Sigma_U}{\Sigma_U^\circ}.$$

Theorem (1.3): Suppose that Postulates I - III hold for an isolated system. Then, there is a function

$$\tilde{U} = \tilde{U}(S, V, N_1, \dots, N_r)$$

and a convex domain of definition

$$\Sigma_S = \left[ [0, \infty) \right]^{r+2} = \underset{(S)}{[0, \infty)} \times \underset{(V)}{[0, \infty)} \times \underset{(N_1)}{[0, \infty)} \times \dots \times \underset{(N_r)}{[0, \infty)}$$

that satisfies

$$\tilde{U}(\tilde{S}(U, V, \vec{N}), V, \vec{N}) = U, \quad \forall (U, V, \vec{N}) \in \Sigma_U$$

and

$$\tilde{S}(\tilde{U}(S, V, \vec{N}), V, \vec{N}) = S, \quad \forall (S, V, \vec{N}) \in \Sigma_S$$

Moreover,  $\tilde{U}$  is a twice continuously differentiable convex function with the property

$$\left(\frac{\partial \tilde{U}}{\partial S}\right)_{V, N_1, \dots, N_r} > 0, \quad \forall (S, V, \vec{N}) \in \Sigma_S.$$

Example (1.4): Suppose that, for a unary material ( $r=1$ ),

$$\tilde{S}(U, V, N) = \left( \frac{NVUR^2}{v_0 \Theta} \right)^{1/3}, \quad \mathcal{Z}_U = [0, \infty)^3,$$

where  $R, v_0, \Theta > 0$  are constants. Consider the function

$$\tilde{U}(S, V, N) = \frac{S^3 v_0 \Theta}{NVR^2}, \quad \mathcal{Z}_S = ?$$

Then

and  $\tilde{S}(\tilde{U}(S, V, N), V, N) = S, \quad \forall (S, V, N) \in \mathcal{Z}_S$

$$\tilde{U}(\tilde{S}(U, V, N), V, N) = U, \quad \forall (U, V, N) \in \mathcal{Z}_U.$$

Defn (1.5): Suppose that Postulates I - III hold. Then the temperature of an equilibrium isolated system is defined as

$$(1.2) \quad T_U(S, V, N_1, \dots, N_r) \equiv \left( \frac{\partial \tilde{U}}{\partial S} \right)_{V, N_1, \dots, N_r}.$$

Postulate IV: Equilibrium of a composite system is that state  $((U^\alpha, V^\alpha, \vec{N}^\alpha), (U^\beta, V^\beta, \vec{N}^\beta)) \in \mathcal{Z}_U^\alpha \times \mathcal{Z}_U^\beta$  such that

$$S = S^\alpha + S^\beta$$

is at its maximum possible value.

Defn (1.6): The function

$$(1.3) \quad P_v(S, V, N_1, \dots, N_r) := - \left( \frac{\partial \tilde{U}}{\partial V} \right)_{S, N_1, \dots, N_r}$$

is called the pressure of a system. The function

$$(1.4) \quad \mu_{v,i}(S, V, N_1, \dots, N_r) := \left( \frac{\partial \tilde{U}}{\partial N_i} \right)_{S, N_j \neq N_i}$$

is called the  $i^{\text{th}}$  chemical potential.

Theorem (1.7): let  $\tilde{S}$  and  $\tilde{U}$  be the entropy and internal energy functions of an isolated system. Then,

$$\frac{\partial \tilde{S}}{\partial U} = \frac{1}{T_s(U, V, \vec{N})},$$

where

$$T_v(S, V, \vec{N}) = T_s(\tilde{U}(S, V, N), V, \vec{N})$$

and

$$T_s(U, V, \vec{N}) = T_v(\tilde{S}(U, V, \vec{N}), V, \vec{N}).$$

Furthermore,

$$\frac{\partial \tilde{S}}{\partial V} = \frac{P_s(U, V, \vec{N})}{T_s(U, V, \vec{N})},$$

where

$$P_v(S, V, \vec{N}) = P_s(\tilde{U}(S, V, N), V, \vec{N}),$$

and

$$P_s(U, V, \vec{N}) = P_v(\tilde{S}(U, V, \vec{N}), V, \vec{N}).$$

And, finally,

$$\frac{\partial \tilde{S}}{\partial N_j} = - \frac{\mu_{s,j}(U, V, \vec{N})}{T_s(U, V, \vec{N})},$$

where

and  $\mu_{u,j}(S, V, \vec{N}) = \mu_{s,j}(\tilde{U}(S, V, \vec{N}), V, \vec{N})$

$$\mu_{s,j}(U, V, N) = \mu_{u,j}(\tilde{S}(U, V, \vec{N}), V, \vec{N}).$$

Proof Exercise. //

Remark: We will usually abuse notation and just write

$$T_u = T_s, \quad P_u = P_s, \quad \mu_{u,j} = \mu_{s,j},$$

when the usage may inferred from the context.

Postulate V: The entropy of an isolated system is zero when the temperature is zero, that is,

$$T_s(U, V, N_1, \dots, N_r) = 0 \Rightarrow \tilde{S}(U, V, N_1, \dots, N_r) = 0.$$

Example (1.8): Consider a unary ( $r=1$ ) isolated system in equilibrium with the fundamental relation

$$\tilde{S}(U, V, N) = \left( \frac{NVUR^2}{v_0 \Theta} \right)^{1/3}.$$

We can set

$$\Sigma_v = [0, \infty)^3.$$

$\tilde{S} = 0$  for all  $(U, V, N) \in \partial \Sigma_v$ , where

$$\partial \Sigma_U = \{(U, V, N) \in \Sigma_U \mid U=0, \text{ or } V=0, \text{ or } N=0\}$$

Recall,

$$T_S(U, V, N) = \frac{3v_0 \theta}{NVR^2} \tilde{S}(U, V, N)$$

$$= \frac{3v_0 \theta}{NVR} \left( \frac{NVUR^2}{v_0 \theta} \right)^{1/3}.$$

$T_S$  is not defined for all of  $\Sigma_U$ ! It is defined for all

$$(U, V, N) \in \text{dom}(T_S) \subset \Sigma_U$$

Define

$$AZ := \{(U, V, N) \in \text{dom}(T_S) \mid T_S(U, V, N) = 0\}.$$

Then

$$AZ \subset \partial \Sigma_U. \quad (\text{Postulate V})$$

I leave it as a homework exercise to determine  $\text{dom}(T_S)$  and  $AZ$ .

Remark: From this point forward, we will assume that all postulates hold.

Theorem (1.9) Suppose that in a composite system  $\alpha$  and  $\beta$  are separated by a diathermal wall. Then, the equilibrium of the composite system may be characterized by

$$(1.5) \quad U^\alpha + U^\beta = U_0$$

and

$$(1.6) \quad T^\alpha(U^\alpha, V^\alpha, \vec{N}^\alpha) = T^\beta(U^\beta, V^\beta, \vec{N}^\beta).$$



Proof: Internal energy may be exchanged between systems  $\alpha$  and  $\beta$  but cannot be exchanged with the outside world. Thus (1.5) must hold because of energy conservation. At equilibrium we must have

$$S(U^\alpha) = S^\alpha(U^\alpha, V^\alpha, \vec{N}^\alpha) + S^\beta(U_0 - U^\alpha, V^\beta, \vec{N}^\beta)$$

and

$$\frac{\partial S}{\partial U^\alpha} = 0.$$

Note, all other variables besides  $U^\alpha$  are fixed.

$$\begin{aligned} 0 = \frac{\partial S}{\partial U^\alpha} &= \frac{\partial S^\alpha}{\partial U^\alpha} + \frac{\partial S^\beta}{\partial U^\beta} \frac{\partial}{\partial U^\alpha} (U_0 - U^\alpha) \\ &= [T^\alpha(U^\alpha, V^\alpha, \vec{N}^\alpha)]^{-1} + [T^\beta(U_0 - U^\alpha, V^\beta, \vec{N}^\beta)]^{-1} (-1) \end{aligned}$$

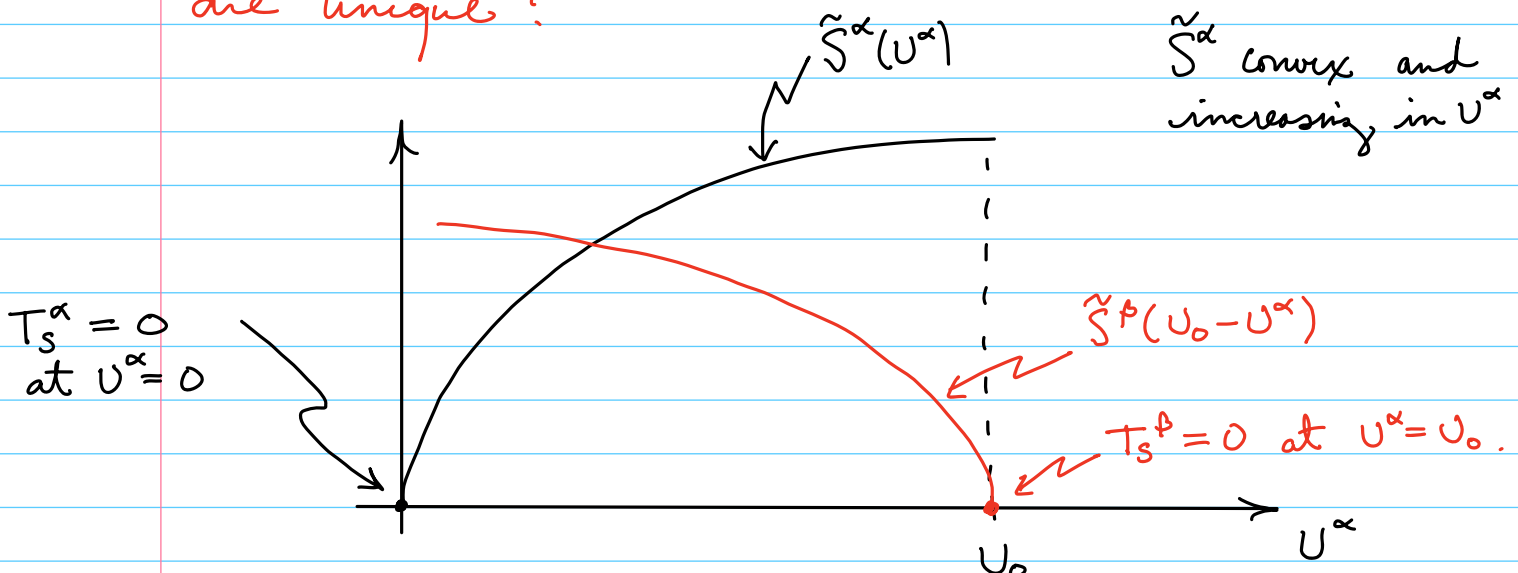
Thus,

$$T^\alpha(U^\alpha, V^\alpha, \vec{N}^\alpha) = T^\beta(U^\beta, V^\beta, \vec{N}^\beta)$$

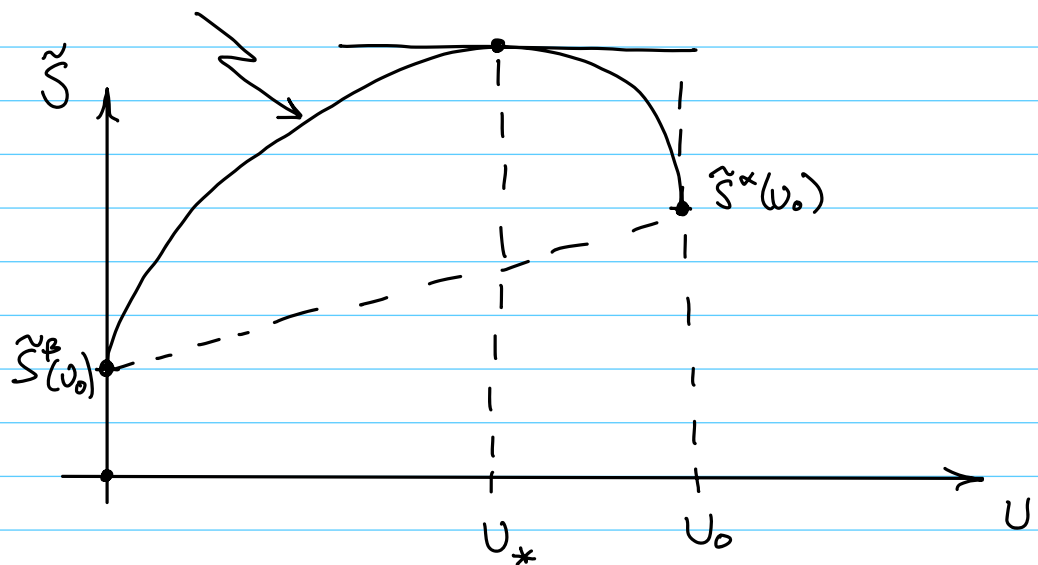
with

$$U^\beta = U_0 - U^\alpha$$

How do we know that solutions exist and are unique?



$$\tilde{S}(U) = \tilde{S}^\alpha(U) + \tilde{S}^\beta(U_0 - U)$$



This proof can also be carried out by using Lagrange multipliers. Set

$$J := S^\alpha(U^\alpha, V^\alpha, \vec{N}^\alpha) + S^\beta(U^\beta, V^\beta, \vec{N}^\beta) + \lambda(U^\alpha + U^\beta - U_0)$$

Then, equilibrium is characterized by

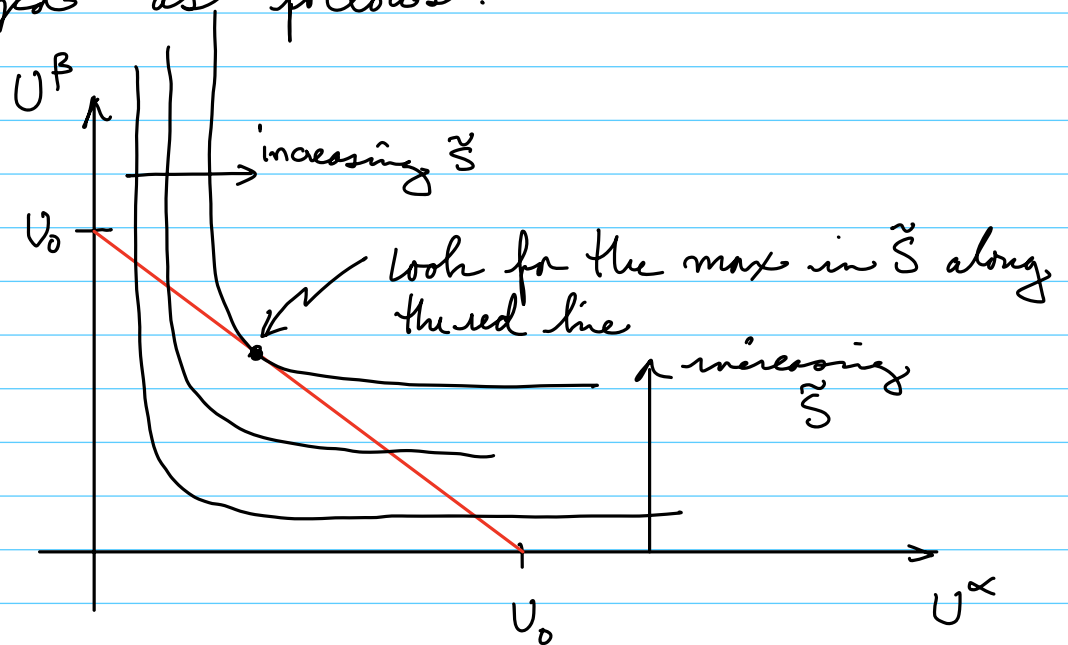
$$0 = \frac{\partial J}{\partial U^\alpha} = \left[ T^\alpha(U^\alpha, V^\alpha, \vec{N}^\alpha) \right]^{-1} + \lambda,$$

$$0 = \frac{\partial J}{\partial U^\beta} = \left[ T^\beta(U^\beta, V^\beta, \vec{N}^\beta) \right]^{-1} + \lambda,$$

$$0 = \frac{\partial J}{\partial \lambda} = U^\alpha + U^\beta - U_0,$$

which yields the same results. ///

The Lagrange Multiplier technique can be visualized as follows:



Theorem (1.9) Suppose that in a composite system  $\alpha$  and  $\beta$  are separated by a diathermal piston. Then, the equilibrium of the composite system may be characterized by

$$(1.7) \quad U^\alpha + U^\beta = U_0,$$

$$(1.8) \quad V^\alpha + V^\beta = V_0,$$

and

$$(1.9) \quad T^\alpha(U^\alpha, V^\alpha, \vec{N}^\alpha) = T^\beta(U^\beta, V^\beta, \vec{N}^\beta), \quad (\text{thermal equil.})$$

$$(1.10) \quad P^\alpha(U^\alpha, V^\alpha, \vec{N}^\alpha) = P^\beta(U^\beta, V^\beta, \vec{N}^\beta). \quad (\text{mechanical equil.})$$

Proof: For this let us use the method of Lagrange multipliers. Note that  $\vec{N}^\alpha$  and  $\vec{N}^\beta$  are fixed. Define

$$\begin{aligned} J := & S^\alpha(U^\alpha, V^\alpha, \vec{N}^\alpha) + S^\beta(U^\beta, V^\beta, \vec{N}^\beta) \\ & + \lambda_U (U^\alpha + U^\beta - U_0) \\ & + \lambda_V (V^\alpha + V^\beta - V_0). \end{aligned}$$

The conditions for equilibrium are

$$0 = \frac{\partial J}{\partial U^\alpha} = \left[ T^\alpha(U^\alpha, V^\alpha, \vec{N}^\alpha) \right]^{-1} + \lambda_U,$$

$$0 = \frac{\partial J}{\partial U^\beta} = \left[ T^\beta(U^\beta, V^\beta, \vec{N}^\beta) \right]^{-1} + \lambda_U,$$

$$0 = \frac{\partial J}{\partial V^\alpha} = \frac{-P^\alpha(U^\alpha, V^\alpha, \vec{N}^\alpha)}{T^\alpha(U^\alpha, V^\alpha, \vec{N}^\alpha)} + \lambda_V,$$

$$0 = \frac{\partial J}{\partial V^\beta} = \frac{-P^\beta(U^\beta, V^\beta, \vec{N}^\beta)}{T^\beta(U^\beta, V^\beta, \vec{N}^\beta)} + \lambda_V.$$

and

$$0 = \frac{\partial J}{\partial \lambda_U} = U^\alpha + U^\beta - U_0,$$

$$0 = \frac{\partial J}{\partial \lambda_V} = V^\alpha + V^\beta - V_0.$$

The result is clear. ///

Here, we have used the fact that

$$\frac{\partial S^\alpha}{\partial U^\alpha} = \frac{1}{T^\alpha}$$

$$\frac{\partial S^\alpha}{\partial V^\alpha} = \frac{P^\alpha}{T^\alpha}$$

and

$$\frac{\partial S^\alpha}{\partial N_i^\alpha} = - \frac{\mu_{i,\alpha}}{T^\alpha}$$

As a short hand, we will write

$$d\tilde{U}^\alpha = T_v^\alpha dS^\alpha - P_v^\alpha dV^\alpha + \sum_{i=1}^r \mu_{v,i}^\alpha dN_i^\alpha$$

and

$$d\tilde{S}^\alpha = \frac{1}{T_s^\alpha} dU^\alpha + \frac{P_s^\alpha}{T_s^\alpha} dV^\alpha - \sum_{i=1}^r \frac{\mu_{s,i}^\alpha}{T_s^\alpha} dN_i^\alpha.$$

Using our abusive notation, we have

$$d\tilde{S}^\alpha = \frac{1}{T^\alpha} dU^\alpha + \frac{P^\alpha}{T^\alpha} dV^\alpha - \sum_{i=1}^r \frac{\tilde{\mu}_i}{T_i} dN_i^\alpha,$$

for example.

Theorem (1.10): Suppose that a composite system is composed of two otherwise isolated systems with no barrier between the systems. Suppose that

$$S^\alpha = S^\alpha(U^\alpha, V^\alpha, \vec{N}^\alpha)$$

and

$$S^\beta = S^\beta(U^\beta, V^\beta, \vec{N}^\beta)$$

are the fundamental entropy relations for the two systems. Then the equilibrium state is defined by the relations

$$(1.11) \quad U^\alpha + U^\beta = U_0$$

$$(1.12) \quad V^\alpha + V^\beta = V_0$$

and

$$(1.13) \quad N_i^\alpha + N_i^\beta = N_{0,i}$$

where  $U_0, V_0, N_{0,i} > 0$ , and

$$(1.14) \quad T^\alpha(U^\alpha, V^\alpha, \vec{N}^\alpha) = T^\beta(U^\beta, V^\beta, \vec{N}^\beta) \quad (\text{thermal equil})$$

$$(1.15) \quad P^\alpha(U^\alpha, V^\alpha, \vec{N}^\alpha) = P^\beta(U^\beta, V^\beta, \vec{N}^\beta) \quad (\text{mech. equil.})$$

$$(1.16) \quad \mu_i^\alpha(U^\alpha, V^\alpha, \vec{N}^\alpha) = \mu_i^\beta(U^\beta, V^\beta, \vec{N}^\beta) \quad (\text{chem. equil.})$$

Proof: The procedure is the same. One can use the method of Lagrange multipliers to do the calculation. In particular, define

$$J := S^\alpha(U^\alpha, V^\alpha, \vec{N}^\alpha) + S^\beta(U^\beta, V^\beta, \vec{N}^\beta) + \lambda_U (U^\alpha + U^\beta - U_0) \\ + \lambda_V (V^\alpha + V^\beta - V_0) + \sum_{i=1}^r \lambda_i (N_i^\alpha + N_i^\beta - N_{0,i}). \quad //$$